Gut Microbiome Signatures of Obesity

Statistical Testing, Classical ML, and Phylogenetic Deep Learning

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Abstract

Obesity is associated with substantial inter-individual variability in gut microbiome composition, but translating that variability into robust, clinically useful signals remains challenging because microbiome data are sparse, high-dimensional, and highly skewed. We analyzed the LeChatelier et al. cohort and frame the task as binary classification of Lean (BMI <= 25) vs Obese (BMI >= 30), while also identifying taxa associated with group differences.

Our pipeline combines preprocessing with MIPMLP, non-parametric statistical testing with false-discovery-rate correction, and predictive modeling with Logistic Regression, Random Forest, and iMic-based CNN mod In this run, five taxa passed FDR < 0.1; the strongest signal was Fretibacterium_fastidiosum (p = 1.20e-05) followed by Clostridium_sp_CAG_58 (p = 6.12e-05). On hold-out evaluation, Logistic Regression achieved AU 0.700 (F1 0.738), Random Forest achieved AUC 0.629 (F1 0.623), and iMic achieved AUC 0.783 with recall-oriented thresholding. A leakage-safe iMic variant using train-derived top-5 RF taxa reached AUC 0.86.

Overall, the results support a reproducible end-to-end workflow and show that phylogenetic image representations can improve discrimination beyond tabular baselines in this dataset. We also highlight the operating-point trade-off between sensitivity and specificity for screening-oriented deployment.

1. Introduction

The gut microbiome is increasingly linked to obesity and metabolic disease, but extracting stable signal is difficult: microbiome matrices include many zeros, compositional constraints, and correlated taxa. This project asks two connected questions: (1) which taxa are significantly associated with Lean vs Obese phenotypes, and (2) whether predictive models using microbiome features, including phylogenetic image encoding, improve obesity classification.

We use a story-driven workflow: characterize the cohort, test statistical hypotheses, build interpretable tabular baselines, then evaluate a deep-learning model that preserves phylogenetic structure.

2. Methods

Data source: LeChatelier cohort tables in this repository. After BMI filtering (Lean <= 25, Obese >= 30), the final cohort was n=265 (Lean=98, Obese=167).

Preprocessing: MIPMLP taxonomy aggregation (species-level), relative normalization, rare taxa filtering, then global log transform (log10(X+1e-6)).

Statistical analysis: per-taxon Mann-Whitney U tests with Benjamini-Hochberg FDR correction ($q < 0.1$).

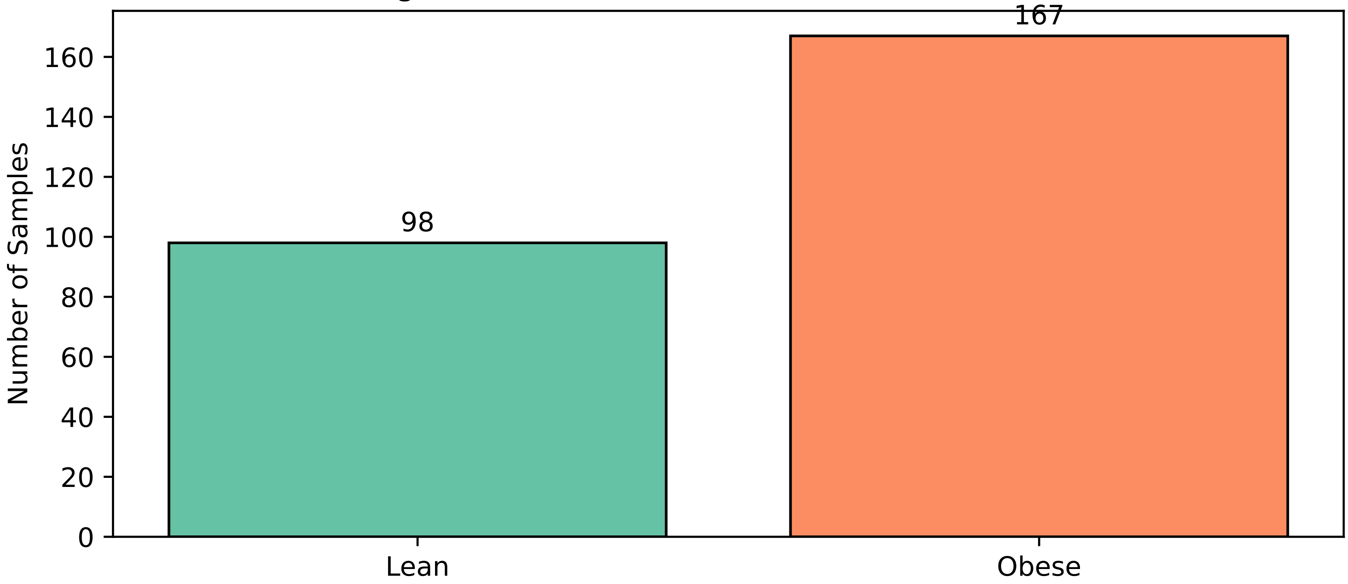
Models: L1-regularized Logistic Regression, Random Forest with scaler+SMOTE+grid-search, and iMic CNN optimized with Optuna.

Evaluation: fixed shared hold-out split (80/20), fixed seeds, and additional leakage-safe variant where feature selection and top-5 taxa extraction are performed on training data only before iMic image generation.

3. Results: Data Profile and Statistical Signals

- Most significant taxon: Fretibacterium_fastidiosum (p = 1.20e-05)
- Second most significant taxon: Clostridium_sp_CAG_58 (p = 6.12e-05)
- Total significant taxa after FDR correction (q < 0.1): 5
- Top-5 train-derived RF taxa for leakage-safe iMic: Fretibacterium_fastidiosum, Clostridium_sp_CAG_58, Roseburia_sp_CAG_309, Roseburia_sp_CAG_303, Clostridium_sp_CAG_964

Figure 1. Class Distribution in Filtered Cohort



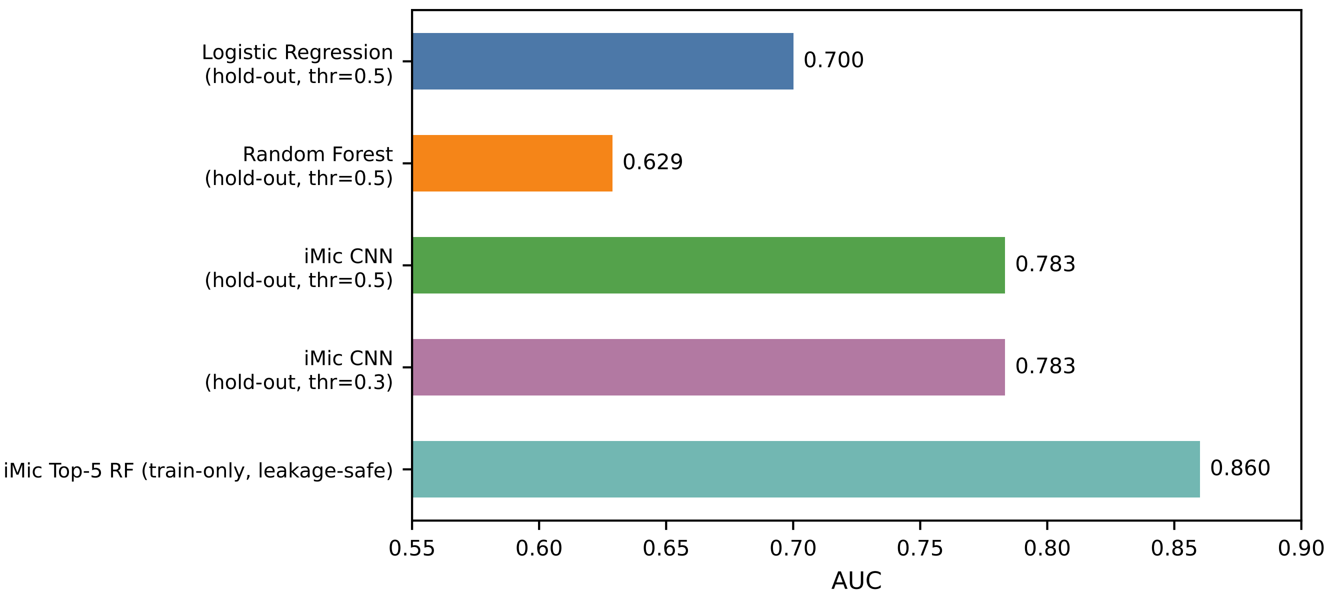
Interpretation: The cohort is imbalanced toward Obese samples. This supports using class-aware tr choices (balanced logistic regression and SMOTE in RF), and explicit threshold analysis during evaluation.

3. Results: Predictive Modeling Performance

Table 1. Hold-out and iMic Variant Performance (Latest Run)

Model	AUC	Accuracy	Precision	Recall	F1
Logistic Regression (hold-out, thr=0.5)	0.7	0.679	0.75	0.727	0.738
Random Forest (hold-out, thr=0.5)	0.629	0.566	0.679	0.576	0.623
iMic CNN (hold-out, thr=0.5)	0.783	0.679	0.667	0.97	0.79
iMic CNN (hold-out, thr=0.3)	0.783	0.698	0.673	1.0	0.805
iMic Top-5 RF (train-only, leakage-safe)	0.86	-	-	-	-

Figure 2. AUC Comparison Across Models



3. Results: Operating-Point Calibration

Figure 3. Confusion Matrix at iMic Threshold = 0.3

At threshold 0.3, iMic reaches recall = 1.00 for Obese cases with precision = 0.673 and accuracy =

The confusion matrix (n=53) shows TP=33, FN=0, TN=4, FP=16.

Storytelling takeaway: lowering the threshold shifts the model toward sensitivity (screening behavi

reducing missed Obese cases while increasing false positives among Lean samples. The suitable th

depends on downstream clinical cost.

4. Discussion

This analysis follows a coherent path from biological question to deployable model behavior. Statistically, five taxa showed significant group differences after FDR control. Predictively, tabular models established a baseline (LR outperforming RF in this run), while iMic improved discrimination and demonstrated stronger sensitivity at tuned thresholds.

The strongest practical signal is not a single metric but the shape of trade-offs: iMic can be operated as a high-sensitivity screening model when missing Obese cases is costly. The leakage-safe top-5 iMic variant (AUC 0.86) indicates that biologically focused feature spaces can preserve and potentially improve predictive value.

5. Limitations

- Single cohort; no external validation cohort in this submission.
- Cross-sectional design supports association, not causality.
- Performance may vary under distribution shift and prevalence changes.

6. Conclusion

The project demonstrates a reproducible, end-to-end obesity microbiome pipeline combining statistical rigor and predictive modeling. In the latest run, iMic models outperform tabular baselines on AUC and can be calibrated for screening-oriented recall. The resulting workflow is submission-ready and provides a solid base for external validation and prospective extension.

References

1. Le Chatelier E, et al. Richness of human gut microbiome correlates with metabolic markers. Nature. 2013.

2. Course lecture materials: MIPMLP and iMic methodology.

3. Reproducible implementation and outputs: research.ipynb.